

PPS 10.1 - Kirchhoff's Second Law, Adding Resistances

1. F: $\varepsilon = V_2 + V_3$ and D: $\varepsilon - V_3 = V_1$?

2.

$$\begin{aligned} R_A &= \left(\frac{1}{P} + \frac{1}{Q} \right)^{-1} + P + Q \\ &= 3Q + \frac{2}{3}Q \\ &= \frac{11}{3}Q \end{aligned}$$

$$\begin{aligned} R_B &= \left(\frac{2}{P} \right)^{-1} + Q + Q \\ &= 3Q \end{aligned}$$

$$\begin{aligned} R_C &= 2 \left(\left(\frac{1}{P} + \frac{1}{Q} \right)^{-1} \right) \\ &= 2 \left(2\frac{Q}{3} \right) \\ &= \frac{4}{3}Q \end{aligned}$$

$$\begin{aligned} R_D &= \left(\frac{P}{2} + \frac{Q}{2} \right) \\ &= \frac{3}{2}Q \end{aligned}$$

So network C has the lowest resistance.

3. C: $\left(\frac{1}{R} + \frac{1}{R} \right)^{-1} = \left(\frac{2}{R} \right)^{-1} = \frac{R}{2}\Omega$

4. A: $\left(\frac{1}{4} + \frac{1}{4} + \frac{1}{4} \right)^{-1} = \left(\frac{3}{4} \right)^{-1} = \frac{4}{3}\Omega$

5. From K_2 , $\varepsilon = V_X + V_X = 2V_X$. The voltage across resistor X is the same as the voltage measured because all resistors have the same resistance (they are identical). Therefore, $V_X = 6.0\text{V}$ so the e.m.f. of the battery is 12.0V . The answer is C (*p.d.* = 6.0V , *e.m.f.* = 12.0V)

6. Assuming all resistors in the circuits have the same resistances:

- a. 6.0V
- b. 3.0V
- c. 12.0V
- d. 12.0V
- e. 3.0V
- f. 4.0V
- g. 2.0V

- 7. a. 10Ω
- b. 5Ω
- c. 8.3Ω
- d. 6.7Ω

8. a. Kirchhoff's first law is that the current going into a junction in a circuit is the same as the currents coming out of the junction. The quantity associated with this law (that is conserved) is current.

Kirchhoff's second law is that the sum of e.m.f.s is equal to the sum of the p.d.s around a closed loop. The quantity that is conserved according to this law is voltage.

- b. i. $0.030 - 0.010 = 0.020\text{A}$
 ii. $750\Omega = \frac{V}{0.020} \cdot V = 750 \times 0.020 = 15\text{V}$
 iii. $R_2 = \frac{15}{0.010} = 1500\Omega$

$$R_1 + \left(\frac{1}{1500} + \frac{1}{750} \right)^{-1} = \frac{45}{0.030}$$

$$R_1 = \frac{45}{0.030} - \frac{1500}{3} = 1000\Omega$$

9. a. $V = 0.3\text{A} \times 4\Omega$
 $= I \times 6\Omega$

$$I = \frac{0.3\text{A} \times 4\Omega}{6\Omega} = 0.20\text{A}$$

- b. From Kirchhoff's first law: $0.20 + 0.30 = 0.50\text{A}$

- c. $\left(\frac{1}{6} + \frac{1}{4} \right)^{-1} + 5.6 = 8\Omega$

Extension Questions

Infinite Resistors

a. $R_n = \frac{1}{2^{n-1}}$

Therefore $R_{137} = \frac{1}{2^{136}} = 1.15 \times 10^{-41}$

b. Let the total resistance in the circuit be equal to R . Then,

$$\begin{aligned} R &= \frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\ \therefore 2R &= 2 + \frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \\ &= 2 + R \end{aligned}$$

So $2R = 2 + R \Rightarrow R = 2\Omega$

c.

$$\begin{aligned} 2\Omega &= \frac{6.0}{I} \\ I &= 3.0\text{A} \end{aligned}$$

$$R_{17} = \frac{1}{2^{16}} = 1.525878906 \times 10^{-5}$$

$$\begin{aligned} V_{17} &= R_{17} \times I \\ &= (1.53 \times 10^{-5}) \times 3.0 \\ &= 4.58 \times 10^{-5}\text{V} \end{aligned}$$

d.

$$\begin{aligned} R &= \frac{1}{2^8} = 0.00390625 \\ P &= I^2 R \\ &= 3^2 \times 0.00390625 = 0.0352\text{W} \end{aligned}$$

e.

$$\begin{aligned} R &= \frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \\ \therefore 2R &= 2 + \frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \dots \\ &= 2 + R \end{aligned}$$

So $2R = 2 + R \Rightarrow R = 2\Omega$

f. $R_{25} = 2^{24} = 1.7 \times 10^7 \Omega$ $V = 6.00\text{V}$ $I = \frac{V}{R} = \frac{6}{1.7 \times 10^7} = 3.53 \times 10^{-7}\text{A}$